

Multiple Choice (10 marks)

1. Which of the following is not a function of vitamin A
 - (a) Synthesis of the blood clotting proteins
 - (b) Prevention of bone loss
 - (c) Regulation of gene expression and cell differentiation
 - (d) Boosting immune system function
2. When using bioelectrical impedance for measuring body composition, why does an electrical current travel faster through muscle than it does through fat?
 - (a) Fat conducts electricity better than muscle
 - (b) Fat is denser than muscle
 - (c) Fat contains less water than muscle
 - (d) Muscle weighs more than fat
 - (e) Muscle contains less water than fat
3. Loss of muscle fiber may be due to loss of neurons which may be due to
 - (a) Exposure to extreme cold
 - (b) Clogged blood vessels
 - (c) Loss of collagen
 - (d) Loss of skin elasticity
 - (e) Decreased water intake
4. Which of the following factors has NOT received as much support as the others in research investigating why social support is related to better health?
 - (a) Physiological
 - (b) Behavioral
 - (c) Biological
 - (d) Anthropological
 - (e) Environmental
5. Which member of the paramyxovirus family can cause very serious croup?
 - (a) Meta pneumo virus
 - (b) Measles
 - (c) Ebola
 - (d) Nipah virus
 - (e) Newcastle disease virus

True / False (10 marks)

Answer True or False

6. True or False: A person weighing 62 kg with a drug dose of 15 mg/kg would require a total dose of 1.86 grams.
7. True or False: The complete oxidation of glucose to carbon dioxide and water yields 45 molecules of ATP in aerobic respiration.
8. True or False: ABCA 1 is not involved in the absorption of cholesterol in the small intestine.
9. True or False: The net production of ATP via substrate-level phosphorylation in glycolysis is 4 from glucose and 2 from glycogen.
10. True or False: Younger workers do not benefit from job training as much as older workers, due to their inherent learning capabilities.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of polyomaviruses in human health, focusing on their pathogenic potential and the reasons they are not associated with any specific diseases.
12. Discuss the diagnostic criteria for osteopenia and osteoporosis in North America and Western Europe, particularly focusing on the implications for women over age 80 with above-average bone mineral density.

End of Paper